

ER Score API — Deployment & Bubble Integration

This is a small, self-contained API that runs the full ER Score calculation (ER-028 through ER-087 — synced directly from the working prototype and verified against its previously-confirmed worked examples). It's built to deploy on Vercel for free and be called from Bubble via the API Connector plugin.

Keeping this in sync: this engine is auto-extracted from `er-score-prototype.jsx` rather than hand-maintained separately, specifically because the two drifted apart before (this file previously stopped at ER-057 while the prototype moved on to ER-087 without anyone noticing). Whenever the prototype's scoring logic changes, re-extract this file from it rather than editing both independently.

Part 1 — Deploy to Vercel

You do not need to know how to code to do this.

1. Create a free account at vercel.com if you don't have one.
2. Create a free account at github.com if you don't have one.
3. Create a new empty repository on GitHub (e.g. `er-score-api`).
4. Upload these three files/folders into that repository, keeping the exact structure:

```
er-score-api/  
  api/  
    _engine.js  
    score.js  
  package.json
```

(GitHub's web interface lets you drag-and-drop files directly — no command line needed.)

5. Go to vercel.com/new, click "Import Project," and select the GitHub repo you just created.
6. Leave all settings at their defaults and click **Deploy**.
7. Vercel will give you a URL like `https://er-score-api-yourname.vercel.app`.

Your live endpoint is now: `https://er-score-api-yourname.vercel.app/api/score`

Test it's working by visiting that URL in a browser — you should see `{"error":"Use POST"}` , which means it's live and correctly rejecting a plain browser visit (it only accepts POST requests with data).

Part 2 — Connect Bubble to it

1. In Bubble, go to the **Plugins** tab and install the official **API Connector** plugin (free).
 2. Add a new API called `ER Score` .
 3. Add a new API call, e.g. `Calculate Score` :
 - Method: `POST`
 - URL: `https://er-score-api-yourname.vercel.app/api/score`
 - Body type: `JSON`
 - In the body, add the fields listed below, using Bubble's `<field_name>` syntax so they map to your app's actual input fields (e.g. `<age>` , `<annualIncome>`).
 4. Click "Initialize call" with some test values filled in — Bubble will auto-detect the response shape (total score, band, category breakdowns, etc.) so you can reference them in workflows and page elements afterward.
 5. In your Calculator page's workflow (e.g. "when Calculate button is clicked"), add a step to trigger the `Calculate Score` API call, passing in whatever the user just entered into your form.
 6. Display the results by referencing `Result of step X's total` , `Result of step X's band's label` , etc. on your Results page.
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Request Body Fields

Every field below is a number, unless noted otherwise. Only send the fields relevant to the categoryId being scored — extras are ignored, but required ones must be present or the math will be wrong (not an error — it'll just silently treat missing numbers as 0).

Always required

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Field	Type	Notes
categoryId	text	One of: vehicle , boat , rv , motorcycle , trailer , electronics , furniture , vacation , wedding , education , primary_residence , investment_property , renting , other
age	number	
targetRetirementAge	number	Defaults to 67 if omitted
annualIncome	number	
takeHome	number	Monthly take-home income
livingExpenses	number	Monthly, before this purchase
savings	number	
taxableInvestments	number	
retirementBalance	number	
delinquent	true/false	
purchasePrice	number	Not required for renting or education
payMethod	text	cash , finance , lease , credit_full , credit_carry

If financing a standard purchase (payMethod: "finance")

cashDown , hasTradeIn (true/false), tradeInEquity , termMonths , interestRate ,
loanType (fixed / adjustable / balloon / interest_only)

If payMethod: "credit_carry"

minPayment

If categoryId: "investment_property"

estRent , rentalExpenses , ipMortgage

If categoryId: "primary_residence" or "renting" (Housing Situation, ER-051)

housingSituation — one of renting , own_selling , own_keeping , none Then, depending

on that value: `currentMonthlyRent`, `currentHousingPayment`, `saleSalePrice`,
`saleMortgageBalance`, `realtorPct`, `sellingClosingPct`, and (`primary_residence` only)
`newPITI`

If `categoryId`: "renting" (ER-047)

`monthlyRent`, `securityDeposit`, `lastMonthRequired` (true/false)

If `categoryId`: "education" (ER-052)

`eduLoanAmount`, `eduInterestRate`, `eduTermMonths`, `hasDeferment` (true/false),
`defermentMonths`, `expectedIncomeAfterCompletion`

If `payMethod`: "lease" (ER-048/049/050)

`negotiatedVehiclePrice`, `capCostReduction`, `monthlyLeasePayment`, `leaseTermMonths`,
`isSinglePayLease` (true/false), `totalLeaseCost`

Response Shape

```
{
  "total": 82,
  "band": { "label": "Strong Financial Position", "color": "#6FA37A" },
  "overridden": false,
  "overrides": [],
  "cashFlow": { "points": 35, "max": 40, "margin": 2100, "pct": 0.34 },
  "financialCushion": { "points": 30, "max": 35, "savings": 22, "investments": 5,
  "financingQuality": { "points": 17, "max": 25, "breakdown": { "ls": 6, "dp": 4,
  "recommendations": [ { "label": "...", "detail": "...", "delta": 5 } ],
  "raw": { ... full internal detail, rarely needed ... }
}
```

Updating the Math Later

If you make another scoring decision (ER-058 and beyond), only `_engine.js` needs to change. Edit it, push the change to your GitHub repo, and Vercel will automatically redeploy — Bubble doesn't need to be touched, since it's just calling the same URL.